Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
8	*(a)*	(i)	Relative velocity (speed) <u>after</u> a collision = (1) <u>0.55</u> × relative velocity (speed) <u>before</u> a collision (1)	2			2		
		(ii)	At 0 °C e = 0.55 × 0.70 or equivalent (gives 0.39) (1) Use of $e=\sqrt{\frac{h}{H}}$ (1) Rearrangement $h=e^2 H$ Bounce height at 0 °C = 0.074 [m]	1	1 1 1		4	3	
	(b)	(i)	Using $F=\frac{mv-mu}{t}$ (1) F = 211 [N] (1)	1	1		2	1	
		(ii)	34 sin 8° [4.7 m s^{-1}] as vertical velocity (1) Vertical height calculated correctly = 1.14 [m] (1) No – never goes above 1.20 [m] (1)			3	3	1	
		(iii)	Angular velocity = 88 [rad s^{-1}] (1) Moment of inertia = 1.23 × 10^{-4} [kg m^2] (1) Rotational kinetic energy = 0.48 [J] (1)		3		3	3	
		(iv)	At maximum height; there is also linear kinetic energy as well as rotational (1) So Wayne is incorrect - total kinetic energy is higher (1)			2	2		
		(v)	Using Bernoulli equation $p=p_0-\frac{1}{2}\rho v^2$ (1) Realising difference in pressure = $\frac{1}{2}\rho\left(v_1^2-v_2^2\right)$ (1) [= 87.1 Pa] Force = 87.1 × area = 0.37 [N] (1) Comparing with weight of 1.7 N allow **ecf** (1)	1 1	1 1		4	2	
			Question 8 total	**6**	**9**	**5**	**20**	**10**	**0**